

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (IT/CE)

Sem - 3

University Examination, 2010-11

MA 201 Discrete Mathematics

Date: 16.05.11, Monday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks

(4) Don't write anything on the question paper.

Section : 1

Q. 1

Check whether the following are true or false:

07

- Suppose that the universe of discourse of the propositional function $P(x)$ consists of the integers 1,2,3. Then the statement $\forall x((x \neq 3) \rightarrow p(x))$ is equal to $P(1) \vee P(2)$.
- Let p and q be two propositions. If p is false then $p \rightarrow q$ is false.
- Every distributive lattice may not be a chain.
- Every element of a distributive lattice has at least one complement.
- Let $(B, *, \oplus, ', 0, 1)$ be a boolean algebra with $B = \{x_1, x_2\}$. Then $(x_1 * x_2)' \oplus x_1 * (0' + 1)$ is a boolean expression.
- With two variables x_1, x_2 , the minterm of $x_1 * x_2'$ is m_3 .
- $(p \oplus q) * (p \oplus q') = p$

Q. 2

(a) Construct a truth table for the following compound proposition:

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$$(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$$

(b) Define: reflexive relation, symmetric relation. Check whether the following relation R is a POSet or not.

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$$R = \{(x, y) : x, y \in \mathbb{Z} \text{ \& } x - y \text{ is even}\}$$

(c) Define: Boolean algebra. Prove that $(S_{30}, *, \oplus, ', 0, 1)$ is a Boolean algebra.

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OR

Q. 2

(a) Let p and q be the propositions.

4

p : You drive over 65 miles per hour.

q : You get a speeding ticket.

Write the following propositions using p and q and logical connectives,

(i) You do not drive over 65 miles per hour.

(ii) You drive over 65 miles per hour, but you do not get a speeding ticket.

(iii) If you do not drive over 65 miles per hour then you will not get a speeding ticket.

(b) Show that $(\mathbb{Z}^+, /)$ is a POSet.

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(c) If $(B, *, \oplus, ', 0, 1)$ is a Boolean algebra and $a, b \in B$ then prove that

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$$a \leq b \Leftrightarrow a * b' = 0 \Leftrightarrow b' \leq a' \Leftrightarrow a' \oplus b = 1$$

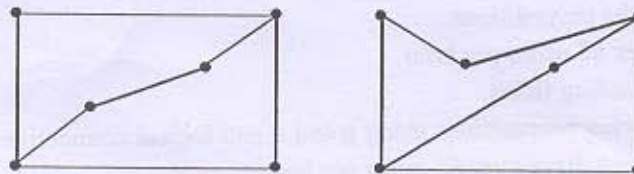
- Q. 3 (a) Define: tautology. Prove that $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology. 4
- (b) Explain the hasse diagram. Draw the hasse diagram of $(S_{100}, /)$. 5
- (c) Define: minterm, maxterm. Express $x_1 + x_2$ as the sum of the product canonical form. 5

OR

- Q. 3 (a) Show that the following are logically equivalent: 4
- (i) $p \leftrightarrow q \& (p \rightarrow q) \wedge (q \rightarrow p)$ (ii) $\neg(p \oplus q) \& p \leftrightarrow q$.
- (b) Define: lattice. Prove that the direct product of two lattices is a lattice. 5
- (c) Prove that (i) $(a' * b' * c) \oplus (a * b' * c) \oplus (a * b' * c') = b' * (a \oplus c)$ 5
- (ii) $(a * b) \oplus (a * b') \oplus (a' * b) \oplus (a' * b') = 1$

Section: 2

- Q. 4 Check whether the following are true or false: 07
1. A complete graph K_3 has 4 spanning trees.
 2. Kuratowski's 1st graph is a planar graph.
 3. Let $T=(V,E)$; $V=\{a,b,c\}$ be a tree such that $d(a)=1$, $d(b)=2$, $d(c)=1$. Then T is a binary tree.
 4. Let G be any graph with n vertices, e edges. r and μ are the rank and the nullity of the graph respectively. Then $r + \mu = n$.
 5. If a connected graph has 3 vertices with odd degree. Then it has a Euler path.
 6. In a weighted graph, dijkstras algorithm gives the minimum weight between any source to any vertex.
 7. In a binary tree there exists two vertices having degree two.
- Q. 5 (a) Define: odd vertex and even vertex in a graph. Prove that the number of vertices of odd degree is always even. 5
- (b) (i) Define isomorphism between two graphs. Prove that the following graphs are isomorphic: 3



- (ii) In a graph K_4 , find a spanning tree. Also find a fundamental cut set With respect to that spanning tree. 2
- (c) Draw Kuratowski's graphs and write their adjacency matrices. 4

OR

- Q. 5 (a) How many subgraphs with at least one vertex does K_4 have? Draw all possible subgraphs. 5

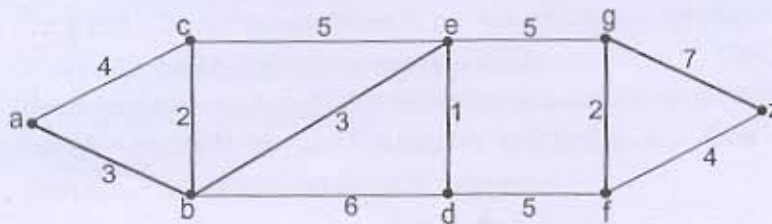
- (b) Define: incidence matrix, adjacency matrix. Draw graph of the following adjacency matrices: 5

$$(i) \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

- (c) Define: connected graph. Find the connected component of the following graph: 4

$G=(V, E)$ where, $V=\{1, 2, 3, 4, 5, 6\}$ & $E=\{(1, 5), (2, 3), (6, 5), (4, 6)\}$.

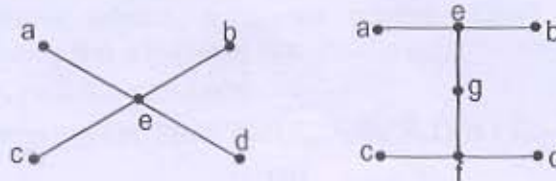
- Q. 6 (a) (i) Define: Eulerian graph, Hamiltonian graph. Give an example of a Eulerian graph which is not Hamiltonian. 3
(ii) Suppose a graph has vertices of degrees 0, 2, 2, 3 and 9. How many edges does the graph have? 2
(b) Find a shortest path between a and z of the following weighted graph 5



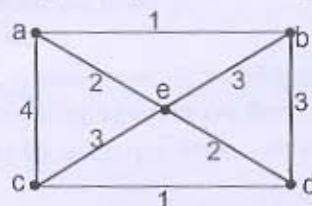
- (c) Define: planar graph. Can we represent the graph K_5 into planar graph? Explain. 4

OR

- Q. 6 (a) Find the union, intersection and ring sum of the given pair of graphs: 5



- (b) Find the minimal spanning tree by Kruskal's algorithm of the following weighted graph: 5



- (c) A simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges. 4